

Effects of Automation and Artificial Intelligence on Employment

Contemporary Economic Issues: ECON 204

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Automation and AI tend to disproportionately affect different types of jobs, leading to labor market polarization.

High-Skilled Jobs:

- Automation complements tasks that require advanced cognitive, analytical, and creative skills (e.g., software development, data analysis).
- Workers in these areas often see increased demand and higher wages due to AI-enabled productivity boosts.

Labor Market Polarization

Automation and AI tend to disproportionately affect different types of jobs, leading to labor market polarization.

Low-Skilled Jobs:

- Certain low-skilled, non-automatable jobs (e.g., caregiving, hospitality) may also remain in demand, particularly where human interaction is essential.

Middle-Skilled Jobs:

- Routine, middle-skill tasks (e.g., clerical work, assembly line operations) are most vulnerable to displacement by AI and robotics, leading to a decline in middle-skill job opportunities.

Job Destruction:

- Automation replaces tasks that are repetitive, routine, and predictable (e.g., manufacturing, bookkeeping, transportation).
- A 2017 study by Frey and Osborne estimated that about 47% of U.S. jobs are at high risk of automation.

Job Creation:

- AI and automation also give rise to new industries, occupations, and tasks (e.g., data scientists, AI specialists, robotics engineers).
- Expansion in sectors like renewable energy, healthcare, and AI-driven tech startups.

Net Effect:

- In the short term, job destruction often outpaces job creation, causing unemployment and worker displacement.
- In the long term, complementary industries and tasks may emerge, but retraining and education are critical to ensuring workers can transition to these roles.

Changes in Job Quality and Nature of Work

AI and automation influence the quality and structure of jobs in several ways:

- **Task Transformation:** Jobs are increasingly divided into tasks, and only certain tasks within jobs are automated.
- **Increased Productivity:** Automation boosts efficiency and reduces costs, potentially leading to higher economic output and growth.
- **Work Intensification:** Workers in automated environments may experience greater performance monitoring and pressure, raising concerns about job satisfaction and mental health.

- **Upward Pressure on High-Skilled Wages:** High-skilled workers who complement automation technologies experience wage gains due to increased productivity and demand.
- **Downward Pressure on Routine Work:** Workers performing automatable tasks often face wage stagnation or reductions due to reduced demand and increased labor supply competition.
- **Bargaining Power:** Automation may weaken the bargaining power of workers in heavily automated sectors.

Skill Requirements and Education

Automation drives a shift in the skills demanded by employers:

- **Reskilling and Upskilling:** Workers need to develop skills such as:
 - Problem-solving, critical thinking, and creativity.
 - Digital literacy and technical expertise.
 - Soft skills like emotional intelligence and communication, particularly for non-automatable jobs.
- **Lifelong Learning:** Traditional education systems must evolve to emphasize continuous learning.

The effects of automation vary across industries:

- **Manufacturing and Logistics:** Robotics and AI are transforming production lines, reducing the need for manual labor.
- **Service Sector:** Automation in customer service and AI in healthcare is augmenting rather than replacing workers.
- **Creative Industries:** AI is encroaching on creative tasks but human input remains critical for originality.

Regional and Demographic Disparities

The impacts of automation are unevenly distributed:

- **Regional Effects:** Regions dependent on manufacturing face greater job losses, while urban areas are more resilient.
- **Demographic Effects:** Younger workers may adapt more easily to technological changes, while older workers may face significant challenges in transitioning to new roles.

Labor economists emphasize the need for proactive policies:

- **Active Labor Market Policies (ALMPs):** Subsidies for retraining programs, wage insurance, job placement services.
- **Social Protection:** Strengthening unemployment insurance and exploring Universal Basic Income (UBI).
- **Regulating Automation:** Encouraging responsible adoption of AI and ensuring ethical AI usage to avoid biases in hiring.

Long-Term Perspectives

Economic Growth: Automation could lead to productivity gains, economic expansion, and higher standards of living.

Redefining Work: A cultural shift may redefine the role of work in society, focusing on creative and fulfilling roles.

However,

- the trajectory of automation's effects depends on technological advancement, worker adaptability, and policy interventions.
- a balanced approach is necessary to ensure inclusive and equitable labor market outcomes.

Problem 1: Labor Market Polarization

Scenario: Automation has affected 3 categories of workers in a country:

- High-skilled workers have seen a 15% increase in demand.
- Low-skilled workers have experienced no change in demand.
- Middle-skilled workers have seen a 25% decrease in demand.

Assume that the total labor force consists of 100,000 workers:

- 20% are high-skilled,
- 50% are middle-skilled,
- 30% are low-skilled.

Question:

- Calculate the total change in labor demand in terms of workers for each skill group.
- What is the net effect on the labor market's composition?

Problem 1: Labor Market Polarization - Solution

Given:

- Total workforce = 100,000
- High-skilled workers = 20% of 100,000 = 20,000
- Middle-skilled workers = 50% of 100,000 = 50,000
- Low-skilled workers = 30% of 100,000 = 30,000
- Changes in demand:
 - High-skilled: +15%
 - Middle-skilled: -25%
 - Low-skilled: 0%

Problem 1: Labor Market Polarization - Solution

Solution:

- High-skilled change in demand: $20,000 \times 0.15 = 3,000$
- Middle-skilled change in demand: $50,000 \times (-0.25) = -12,500$
- Low-skilled change in demand: $30,000 \times 0 = 0$
- High-skilled workers after change: $20,000 + 3,000 = 23,000$
- Middle-skilled workers after change: $50,000 - 12,500 = 37,500$
- Low-skilled workers after change: $30,000$

Net Effect:

- High-skilled workers increased by 3,000.
- Middle-skilled workers decreased by 12,500.
- Low-skilled workers remained unchanged.

Problem 2: Job Displacement and Job Creation

Scenario: A country has a total workforce of 10 million workers. A study suggests that 40% of current jobs are highly susceptible to automation, but for every job lost, 0.75 new jobs are created in complementary sectors like AI, data science, and renewable energy.

Question:

- How many jobs are expected to be lost due to automation?
- How many new jobs are expected to be created?
- What is the net effect on employment in the country? Will the country experience a net increase or decrease in jobs?

Problem 2: Job Displacement and Job Creation - Solution

Given:

- Total workforce = 10 million workers
- 40% of jobs are susceptible to automation
- For every job lost, 0.75 new jobs are created.

Solution:

- Jobs displaced = $10,000,000 \times 0.40 = 4,000,000$
- New jobs created = $4,000,000 \times 0.75 = 3,000,000$
- Net job change = $3,000,000 - 4,000,000 = -1,000,000$

Conclusion:

- The country will experience a net decrease of 1 million jobs in the short term.

Problem 3: Wage Effects

Scenario: Consider a country with two sectors:

- The *High-Skilled Sector*: The average wage is \$80,000. Automation increases worker productivity by 20%, leading to a 10% wage increase for high-skilled workers.
- The *Routine Work Sector*: The average wage is \$40,000. Automation reduces the demand for workers, leading to a 5% decrease in wages.

If there are 3 million workers in each sector, calculate the total wage change in each sector.

Question:

- What is the overall effect on total wage payments across both sectors?

Problem 3: Wage Effects - Solution

Given:

- High-Skilled Sector:
 - Average wage = \$80,000
 - Productivity increases by 20%, wage increases by 10%
 - Number of workers = 3 million
- Routine Work Sector:
 - Average wage = \$40,000
 - Wage decreases by 5%
 - Number of workers = 3 million

Problem 3: Wage Effects - Solution

Solution:

- High-Skilled Sector:
 - Wage increase = $80,000 \times 0.10 = 8,000$
 - New wage = $80,000 + 8,000 = 88,000$
 - Total wage payment = $88,000 \times 3,000,000 = 264,000,000,000$
- Routine Work Sector:
 - Wage decrease = $40,000 \times 0.05 = 2,000$
 - New wage = $40,000 - 2,000 = 38,000$
 - Total wage payment = $38,000 \times 3,000,000 = 114,000,000,000$
- Total wage payment =
 $264,000,000,000 + 114,000,000,000 = 378,000,000,000$
- Total wage increase =
 $378,000,000,000 - 360,000,000,000 = 18,000,000,000$

Conclusion:

- The total wage payment increases by \$18 billion across both sectors.

Problem 4: Skill Requirements and Education

Scenario: A country plans to invest in retraining programs for 1 million displaced workers due to automation. The cost of retraining per worker is \$2,000. After retraining, 70% of workers are expected to find new jobs at a wage of \$50,000 per year, while the remaining 30% remain unemployed.

Question:

- What is the total cost of the retraining program?
- What is the expected annual increase in wage payments for the retrained workers compared to their previous wage of \$30,000?

Problem 4: Skill Requirements and Education - Solution

Given:

- Total displaced workers = 1 million
- Cost of retraining per worker = \$2,000
- 70% of retrained workers will find new jobs at \$50,000 per year
- 30% will remain unemployed
- Previous wage of displaced workers = \$30,000

Problem 4: Skill Requirements and Education - Solution

Solution:

- Total cost of retraining program =
 $1,000,000 \times 2,000 = 2,000,000,000$ (i.e., \$2 billion)
- Number of retrained workers finding new jobs =
 $1,000,000 \times 0.70 = 700,000$
- Wage increase = $50,000 - 30,000 = 20,000$
- Total annual wage increase = $700,000 \times 20,000 = 14,000,000,000$

Conclusion:

- The total cost of the retraining program is \$2 billion.
- The expected annual increase in wage payments for retrained workers is \$14 billion.

Problem 5: Regional and Demographic Disparities

Scenario: The automation of manufacturing has led to job displacement in two regions:

- Region A (heavy reliance on manufacturing): 40% of workers are displaced due to automation.
- Region B (diversified economy): 20% of workers are displaced.

The total workforce in Region A is 3 million workers, and in Region B is 5 million workers. Additionally, 30% of displaced workers are aged 50 and above, while the remaining are under 50.

Problem 5: Regional and Demographic Disparities

Question:

- Calculate the total number of displaced workers in each region.
- In Region A, if only 50% of displaced workers under 50 years old find new employment, how many workers will remain unemployed in Region A?
- What percentage of displaced workers in both regions are aged 50 and above?

Problem 5: Regional and Demographic Disparities - Solution

Given:

- Region A: Total workforce = 3 million, 40% displaced.
- Region B: Total workforce = 5 million, 20% displaced.
- 30% of displaced workers are aged 50 and above, while the remaining are under 50.

Problem 5: Regional and Demographic Disparities - Solution

Solution:

- Total displaced workers:
 - Region A: $3,000,000 \times 0.40 = 1,200,000$
 - Region B: $5,000,000 \times 0.20 = 1,000,000$
- Displaced workers aged 50 and above:
 - Region A: $1,200,000 \times 0.30 = 360,000$
 - Region B: $1,000,000 \times 0.30 = 300,000$
- Displaced workers aged under 50:
 - Region A: $1,200,000 - 360,000 = 840,000$
 - Region B: $1,000,000 - 300,000 = 700,000$
- Region A unemployment (under 50): $840,000 \times 0.50 = 420,000$

Problem 5: Regional and Demographic Disparities - Solution

Conclusion:

- Total displaced workers: 1,200,000 in Region A and 1,000,000 in Region B.
- Displaced workers aged 50 and above: 360,000 in Region A and 300,000 in Region B.
- Unemployed in Region A (under 50): 420,000 workers.